

MUGODZHARSKOYE, KazakhstanWMO: **355320**Lat: **48.5869N**Lon: **58.4667E**Elev: **1306**StdP: **14.02**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-9.8	-21.1	1.8	-14.2	-16.5	2.3	-9.6	51.3	-7.5	45.0	-1.1	11.5	230	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	94.8	65.4	91.6	64.3	88.7	63.3	67.2	88.0	65.7	86.8	64.5	85.1	9.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	82.8	73.6	58.1	75.8	73.0	55.8	69.6	73.2	32.3	87.5	31.2	87.1	30.2	85.2	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.1	31.3	26.6	-20.4	98.9	7.4	3.3	-25.8	101.3	-30.1	103.3	-34.2	105.1	-39.6	107.5		
(5)			-20.2	70.3	7.1	2.4	-25.4	72.0	-29.5	73.5	-33.5	74.8	-38.7	76.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	10.1	11.9	25.6	46.9	60.8	71.6	74.7	72.9	59.9	44.5	28.2	16.0
	DBStd	25.36	10.56	11.11	10.70	10.83	8.65	7.77	6.95	8.11	9.14	9.45	10.25	10.50
	HDD50	5212	1238	1068	757	181	13	0	0	0	21	225	656	1053
	HDD65	8671	1703	1488	1221	547	188	28	9	26	201	637	1105	1518
	CDD50	2934	0	0	2	90	346	648	766	710	319	53	0	0
	CDD65	917	0	0	0	5	56	226	310	271	48	1	0	0
	CDH74	9819	0	0	0	43	534	2417	3404	2962	453	6	0	0
	CDH80	4101	0	0	0	2	136	1014	1512	1300	137	0	0	0
	WSAvg	10.0	12.8	13.0	12.2	9.9	9.5	9.0	8.5	8.1	8.1	8.9	9.3	11.3
	PrecAvg	10.3	0.8	0.7	0.8	0.9	1.2	1.1	1.0	0.6	0.4	1.0	1.0	0.9
(15) Precipitation	PrecMax	15.7	1.5	1.4	1.9	1.9	2.5	4.3	3.3	1.3	0.9	1.7	1.9	2.5
	PrecMin	5.9	0.2	0.2	0.1	0.3	0.0	0.3	0.1	0.1	0.0	0.3	0.2	0.3
	PrecStd	2.2	0.3	0.3	0.5	0.5	0.6	0.9	0.8	0.4	0.3	0.4	0.4	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	35.9	58.5	79.1	87.5	97.0	98.3	97.9	91.7	72.9	53.2	39.8
		MCWB	31.8	32.3	46.8	58.5	61.8	65.4	66.3	66.5	63.9	54.8	45.4	35.2
	2%	DB	30.7	32.1	49.6	73.6	83.9	93.1	94.5	94.1	83.5	68.1	48.1	34.5
		MCWB	29.3	29.9	42.0	56.2	61.0	64.3	65.7	65.1	60.2	53.3	42.2	31.1
	5%	DB	28.2	29.6	43.7	68.4	80.2	89.6	91.3	90.6	79.1	63.8	44.3	32.0
		MCWB	26.7	27.6	38.5	53.4	60.0	63.3	64.4	63.9	58.8	50.7	39.6	30.0
	10%	DB	25.1	27.1	39.2	63.6	76.0	86.1	88.2	87.2	75.1	59.4	40.8	29.2
		MCWB	23.5	25.2	35.0	51.3	58.2	62.3	63.4	62.9	57.0	48.3	36.9	27.1
	0.4%	WB	32.2	32.8	47.2	60.0	64.5	68.3	69.8	69.0	65.1	57.1	47.5	35.0
		MCDB	34.2	34.9	56.8	77.0	82.2	88.4	89.2	89.3	87.9	69.3	50.4	38.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.5	29.9	42.3	56.8	62.3	65.9	67.3	66.7	61.4	53.8	42.8	31.9
		MCDB	30.6	31.8	48.8	71.1	80.1	88.1	87.9	88.3	81.2	66.5	47.6	33.6
	5%	WB	26.7	27.8	38.6	54.2	60.5	64.4	65.9	65.2	59.3	51.3	40.1	29.7
		MCDB	28.1	29.6	43.6	67.0	77.5	85.7	86.2	87.0	77.2	62.4	44.1	32.1
	10%	WB	23.6	25.2	35.2	51.7	58.7	62.9	64.7	63.7	57.4	48.7	37.0	27.2
		MCDB	25.1	27.1	38.9	63.0	74.9	83.5	84.4	85.3	73.9	59.0	40.8	29.2
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	12.1	12.4	17.3	19.2	19.9	19.2	20.0	19.5	16.8	10.9	10.8
		MCDBR	10.6	11.3	17.5	22.6	22.4	23.3	22.2	22.3	23.1	22.1	14.2	11.1
		MCWBR	10.0	10.3	12.9	12.4	9.9	8.6	7.9	8.0	10.2	12.8	10.8	9.8
	5% WB	MCDBR	9.6	11.1	17.0	21.5	20.9	21.2	20.6	21.6	22.3	21.2	13.2	10.8
		MCWBR	9.6	10.3	12.7	13.4	10.5	9.4	8.9	9.1	10.9	13.3	10.9	9.8
(40) Clear-Sky Solar Irradiance	taub		0.295	0.320	0.384	0.426	0.417	0.411	0.404	0.410	0.358	0.350	0.329	0.302
	taud		2.113	2.043	2.024	2.071	2.177	2.254	2.288	2.250	2.353	2.289	2.236	2.144
	Ebn at Noon		234	256	257	260	268	270	270	262	267	246	221	213
	Edh at Noon		27	37	45	48	45	42	40	40	32	29	24	23
(44) All-Sky Solar Radiation	RadAvg		471	843	1194	1564	1998	2186	2103	1844	1418	830	411	308
	RadStd		74	78	133	159	155	126	104	115	75	89	44	55

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.17	N/A	N/A	+2.27	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+1.17	N/A	N/A	+2.27	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page